Name _____ Period _____ Date _____

Phenomena: The Siamese Cat Mystery

Focus question: How can a difference in DNA change which parts of a cat are dark?

I notice…	I wonder…

Part 1: My initial model

Draw what is inside a Siamese cat's cells that leads to dark ears, paws, and tail.

Part 2: Key ideas

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Phenomena: The Siamese Cat Mystery (continued)

Part 3: Temperature and fur color

Body part	Skin temp (°C)	Fur color
Back		
Belly		
Upper leg		
Ear		
Nose and face		
Paw		
Tail tip		

Pattern: dark fur goes with _____ skin.

Part 4: Litters of kittens

Parents	What kittens did they have?
Siamese × Siamese	
Siamese × solid	
Solid × solid	

Exit ticket

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What Is DNA Made Of?

Focus question: What is DNA built from, and what is the message?

Warm-up

Sketch what you think DNA looks like.

Part 1: One nucleotide

Draw and label the sugar, phosphate, and base.

Part 2: Words to know

Word	In my words
double helix	
nucleotide	
backbone	
base pair	

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What Is DNA Made Of? (continued)

Part 3: Partner strands

Strand	Partner strand
A T G C C T	
T T T G A C	
C T G G G A	

The order of the bases is the _____.

Part 4: One letter apart

Solid cat: A T G G G A C T G T T T
Siamese cat: A T G A G A C T G T T T

How many letters differ? Where?

Exit ticket

Name _____ Period _____ Date _____

Build, Copy, and Read a Gene

Focus question: How does a gene's message become a chain of amino acids?

Warm-up

In my own words, what does “copy, then read” mean?

Part 1: Lab record

Step	What I built (letters or colors)
A: Template strand	
B: Partner strand	
C: The copy	
C: Amino acid chain	

Which strand carries the “coding” letters? How do you know?

Name _____ Period _____ Date _____

Build, Copy, and Read a Gene (continued)

Part 2: Change one letter

Step	What I found
Changed template	
New copy	
New amino acid chain	

How many amino acids changed? Which one?

Part 3: Connect to the cat

What did one letter do to the chain? Name two ways the gummy model is different from real DNA.

Exit ticket

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What Can One Letter Do?

Focus question: What can a one-letter change in DNA do?

Warm-up

Part 1: Card sort, glue each card under its category

1: Changes an amino acid in the chain	2: Changes whether or when the protein is made	3: No noticeable change

Part 2: My reasons

Card	Category	My reason
A		
B		
C		
D		
E		
F		

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What Can One Letter Do? (continued)

Part 3: The Siamese cat

DNA	Amino acid	Protein	Fur

How does this explain the dark points and the white newborn?

Claim check: “Every change in DNA changes how a cell works.” Use the cards to respond.

Exit ticket

Name _____ Period _____ Date _____

Same DNA, Different Jobs

Focus question: If all cells have the same DNA, why do cells have different jobs?

Warm-up

Part 1: Two kinds of cells

Cell	Genes in cell	Proteins made	Job
P	1, 2, 3, 4	1 and 2	Carries signals
Q	1, 2, 3, 4	3 and 4	Squeezes to move the body

Same DNA? Why different jobs?

Part 2: Cells in a Siamese cat's ear

Cell type	Has the gene?	Makes protein X?	Job
Pigment-making cell			Makes dark color
Skin cell			Covers and protects
Nerve cell			Carries signals

Name _____ Period _____ Date _____

Same DNA, Different Jobs (continued)

Part 3: Where does the changed protein work?

Works only in skin cooler than about 33 °C (a simplified number).

Body part	Skin temp (°C)	Protein works?	Fur color
Ear	28		
Nose	30		
Paw	28		
Tail tip	27		
Upper leg	34		
Back	36		
Belly	36		

Explain it with the words DNA, gene, protein, and job.

Exit ticket

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Case Data

Focus question: How does a DNA change lead to an effect on the body?

Warm-up

The chain for the cat: one-letter DNA change → _____ → _____ → _____

Part 1: Case 1, Siamese and solid cats

	Solid cat	Siamese cat
Gene		
Amino acid chain		
Where the protein works		
Fur color		

Part 2: Case 2, a human gene

Person	Gene	Protein	Symptoms
1			
2			

The chain for Case 2: _____ → _____ → _____ → _____

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Case Data (continued)

Part 3: Compare the cases

How they are alike	How they are different

Part 4: Claim, evidence, reasoning

Claim: what caused the Siamese cat's dark points?

Evidence: at least two numbers from the data.

Reasoning: how does a one-letter DNA change lead to dark color only on the cooler body parts?

Exit ticket

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Review and Back to the Cat

Focus question: How can I explain the Siamese cat with DNA, genes, and proteins?

Warm-up: My CFA scores

CFA	My score	Review done?
LS1-1.1		
LS1-1.2		
LS1-1.3		
LS1-1.4		
LS1-1.5		
LS1-1.6		

Part 1: Vocabulary

Word	In my words
DNA	
nucleotide	
gene	
base sequence	
amino acid	
protein	

Name _____ Period _____ Date _____

Review and Back to the Cat (continued)

Part 2: My revised model

Show the gene, the one-letter change, the copy-then-read path, the amino acid chain, the protein, and the dark points.

Part 3: Write It, two Siamese kittens

Why are the cool-room kitten's ears darker?

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Unit Assessment and Reflection

Focus question: What do I know about how DNA leads to traits?

Warm-up: Before the test

Three things I am sure about:

Part 1: Back to the cat

What I would add to my Day 1 model now.

Part 2: Looking back at my CFAs

CFA	My score	What helped me
LS1-1.1		
LS1-1.2		
LS1-1.3		

Name _____ Period _____ Date _____

Unit Assessment and Reflection (continued)

CFA	My score	What helped me
LS1-1.4		
LS1-1.5		
LS1-1.6		

Part 3: What I will do next time

Exit ticket

One question I still have about DNA and proteins: